COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 010 - Data Structure & Algorithms | **SECTION:** | CPE21S4 |
| **DATE PERFORMED:** | Jul 26, 2026 | **DATE SUBMITTED:** | Aug 5, 2026 |
| **NAME:** | Bal, Kevin Ranz J. | **INSTRUCTOR:** | Engr Jimlord Quejado |

## Hands-on Activity 1.1 Basic C++ Programming

### 1. PROCEDURE OUTPUT

What procedures did you perform?

| **PROCEDURE:** |
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| **Procedure that performed:** |
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| **ILO A** |
| **Task Title & Task Objective** |
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| **Screenshot of the code** | |
| **Output of the code** | |
| **Analysis** | |

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| **ILO B** |
| **Task Title & Task Objective** |
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|---|---|
| **Screenshot of the code** | |
| **Output of the code** | |
| **Analysis** | |

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

**SUPPLEMENTARY:**

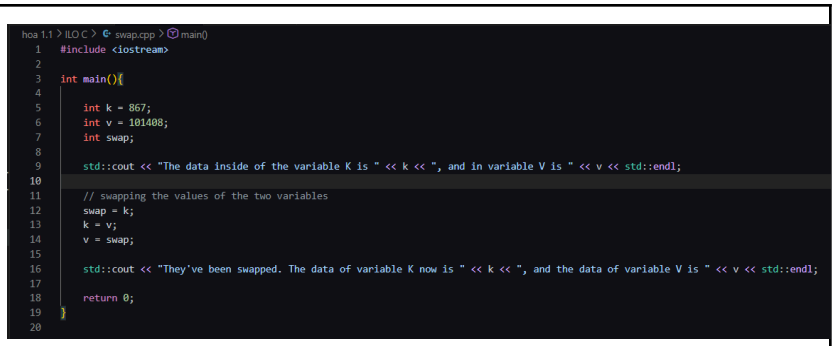
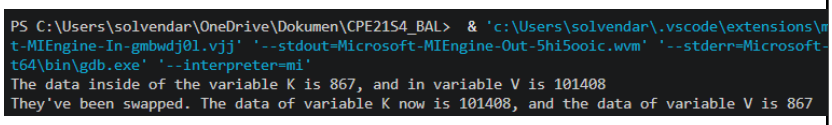
**ILO C**

**Task Title & Objective:**

ILO C: Solve Different Problems using the C++ Programming Language
The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.
1. Create a C++ program to swap the two numbers in different variables.
2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
3. Create a C++ program that has a function that will calculate the distance between two points.

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

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| **1.** | Create a C++ program to swap the two numbers in different variables. |

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| **Screenshot of the code** |  |
| **Output of the code** |  |
| **Analysis** | Code Analysis: I created a program where the data of the variable will swap by starting to declare three integer and the two of it have a data inside while one have not, i put the value of k inside the variable swap while the k value will be inside of the swap and the value i put to the variable of k is v that have a data value too and get the value of swap on the variable of v and the swap was already done. |
| **2.** | Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit. |

```
hoa 1.1 > ILO C > swap.cpp > main()
#include <iostream>

int main(){

    int k = 867;
    int v = 101408;
    int swap;

    std::cout << "The data inside of the variable K is " << k << ", and in variable V is " << v << std::endl;

    // swapping the values of the two variables
    swap = k;
    k = v;
    v = swap;

    std::cout << "They've been swapped. The data of variable K now is " << k << ", and the data of variable V is " << v << std::endl;

    return 0;
}
```

```
PS C:\Users\solvendar\OneDrive\Dokumen\CPE21S4_BAL>  & 'c:\Users\solvendar\.vscode\extensions\m
t-MIEngine-In-gmbwdj0l.vjj' '--stdout=Microsoft-MIEngine-Out-5hi5ooic.wvm' '--stderr=Microsoft-
t64\bin\gdb.exe' '--interpreter=mi'
The data inside of the variable K is 867, and in variable V is 101408
They've been swapped. The data of variable K now is 101408, and the data of variable V is 867
```

| Screenshot of the code | |
|---|---|

Kelvin_to_Fahrenheit.cpp | prototype.cpp

Kelvin_to_Fahrenheit.cpp

```cpp
#include <iostream>

float Fahrenheit(float kelvin);
float Kelvin(float far);

int main()
{

    float conv;
    int choice = 3;

    do
    {
        std::cout << std::endl;
        std::cout << "  Temperature Converter   " << std::endl;
        std::cout << std::endl;
        std::cout << "1. Kelvin to Fahrenheit" << std::endl;
        std::cout << "2. Fahrenheit to Kelvin" << std::endl;
        std::cout << std::endl;
        std::cout << "Input your choice: ";
        std::cin >> choice;

        switch (choice)
        {

        case 1:

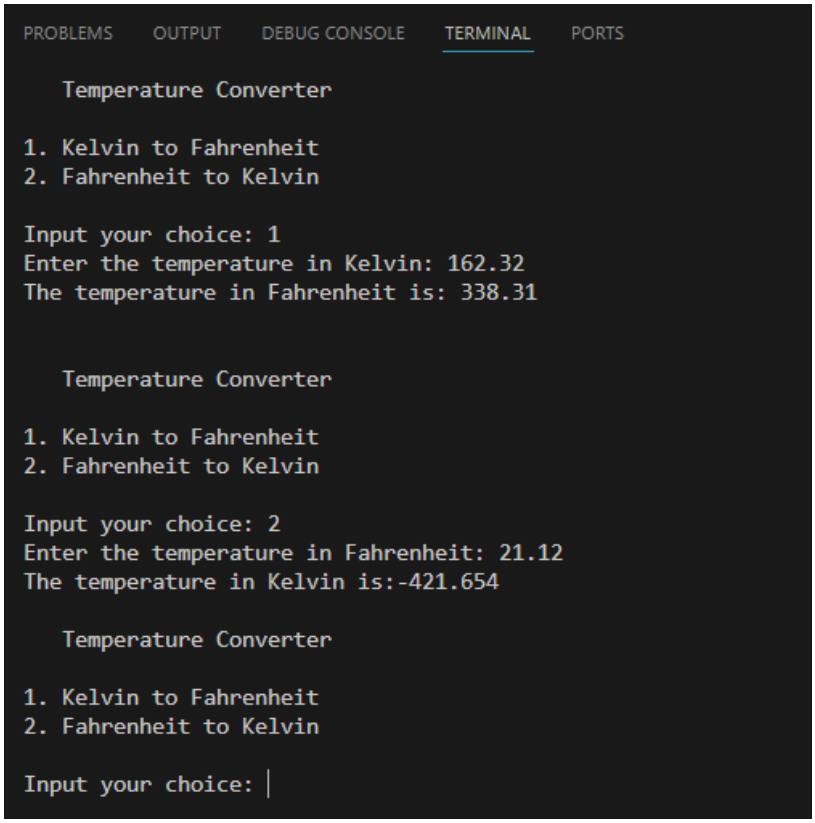
            std::cout << "Enter the temperature in Kelvin:";
            std::cin >> conv;
            std::cout << "The temperature in Fahrenheit is: " << Fahrenheit(conv) << " " << std::endl;
            std::cout << std::endl;
            break;

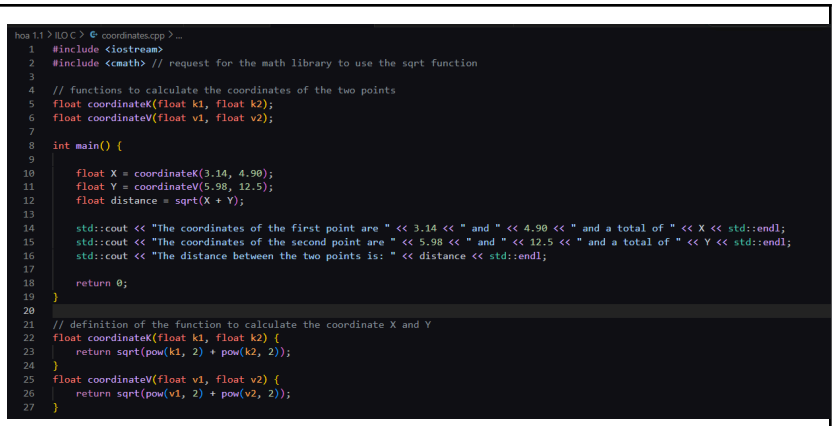
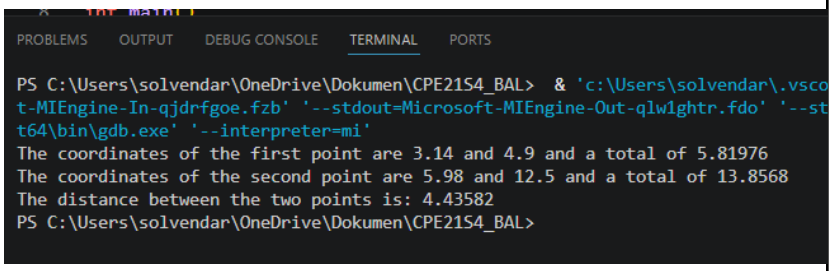
        case 2:

            std::cout << "Enter the temperature in Fahrenheit: ";
            std::cin >> conv;
            std::cout << "The temperature in Kelvin is: " << Kelvin(conv) << std::endl;
            break;

        case 3:
            std::cout << "Invalid choice. Input a valid number." << std::endl;
            break;

        default:
            if (choice !=3){
                std::cout<<"Invalid. Try again later."<<std::endl;
```

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| **Output of the code** |  <br>(This was the old output, I revised it but I can't run it because of the directory. Sending this will be the proof of the program are complete and running) |
| **Analysis** | I created a program where I created two functions with a parameter inside for float and kelvin to compute each of them by putting the formula of kelvin inside my parameter of fahrenheit as my call in method and did the same on the other function. In int main, I declared a variable int choice = 3 and float conv to access the parameters inside of my two functions once I called it out, and I created a Do/While and used switch case and inside was the choice to let the user input its choice what he/she wants to convert and and in the default case, I input the (choice!=3) and stated the choice will become false which will stop the program and after that case I put the while inside and the choice so after the user converted the temperature, it will restart the program. |
| **3.** | Create a C++ program that has a function that will calculate the distance between two points. |

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| **Screenshot of the code** |  |
| **Output of the code** |  |
| **Analysis** | In this program, I declared a <cmath>to access or request for the library so I can use the square root for the coordinates. I declared two functions where the first function the parameter was the x1 and x2 while the 2nd is y1 and y2 that I'll use for the methods and after that, I define the function below the main where i called the first function and return it but in square and using the 1st parameter inside the function and used the pwr for power which is raised to 2 and will be sum to the 2nd parameter of the function and return to get the value of it and did the same to the another function. And inside the main I created a variable float and the data inside it was the function and the parameter inside and same as the Y coordinate and I created a float distance and used the square root and the sum of the coordinate X and Y inside the main and display them both the values inside the parameter of the main, sum of them and the distance that i get after calculating it. |
| **4.** | 4. Modify the code given in ILO B and add the following functions:<br>a. A function to compute for the area of a triangle<br>b. A function to compute for the perimeter of a triangle<br>c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others. |

**Screenshot of the code**

```cpp
#include <iostream>

class Triangle
{

private:
    // for angles
    double totalAngle, angleA, angleB, angleC;

    // for sides, base, and height
    double sideA, sideB, sideC;
    double base, height;

public:
    // revised public constructor to accept the angles, sides, base, and height
    Triangle(double A, double B, double C, double a, double b, double c, double base_val, double height_val);

    void setAngles(double A, double B, double C);
    void setSides(double a, double b, double c, double base_val, double height_val);
    const bool validateTriangle();
    void Area();
    void Perimeter();
    void displayAngles();
    void classifyTriangle();
};

// constructor implementation for the triangle class
Triangle::Triangle(double A, double B, double C, double a, double b, double c, double base_val, double height_val)
{
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;

    sideA = a;
    sideB = b;
    sideC = c;
    base = base_val;
    height = height_val;
}

void Triangle::setAngles(double A, double B, double C)
{

    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
```

```cpp
void Triangle::setAngles(double A, double B, double C)

void Triangle::setSides(double a, double b, double c, double base_val, double height_val)
{
    sideA = a;
    sideB = b;
    sideC = c;
    base = base_val;
    height = height_val;
}

// to compute the area
void Triangle::Area()
{
    double area = 0.5 * base * height;
    std::cout << "Area of the triangle: " << area << std::endl;
}

// to compute the perimeter using side lengths
void Triangle::Perimeter()
{
    double perimeter = sideA + sideB + sideC;
    std::cout << "Perimeter of the triangle: " << perimeter << std::endl;
}

// validation for the triangle base on the total of the angles
const bool Triangle::validateTriangle()
{
    return (totalAngle == 180 && angleA > 0 && angleB > 0 && angleC > 0);
}

// for displaying the angles of the triangle
void Triangle::displayAngles()
{
    std::cout << "Angle A: " << angleA << std::endl;
    std::cout << "Angle B: " << angleB << std::endl;
    std::cout << "Angle C: " << angleC << std::endl;
}

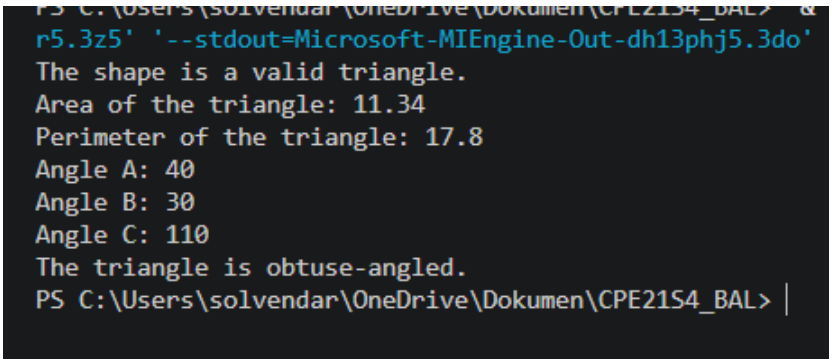
// for classifying the triangle based on its angles
void Triangle::classifyTriangle()
{
    if (angleA < 90 && angleB < 90 && angleC < 90)
    {
        std::cout << "The triangle is acute-angled.\n";
    }
    else if (angleA > 90 || angleB > 90 || angleC > 90)
    {
    }
    else
    {
        std::cout << "The triangle is 'others'\n";
    }
}

int main()
{
    // arguments: (for angles: 40, 30, 110) (for sides: 5.5, 4.2, 8.1) (for base: 8.1, for height: 2.8)
    Triangle set1(40, 30, 110, 5.5, 4.2, 8.1, 8.1, 2.8);

    if (set1.validateTriangle())
    {
        std::cout << "The shape is a valid triangle.\n";
    }
    else
    {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    set1.Area();
    set1.Perimeter();
    set1.displayAngles();
    set1.classifyTriangle();

    return 0;
}
```

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| **Output of the code** |  |
| **Analysis** | In this supplementary last part, I modified the ILO B. I created a prototype for my triangle to keep my data inside clean and organized and to separate the angles to the lengths. While in my private variables, I created a variable where it holds the angles and separated the variables where it holds the sides, base, and height. I made it this way so the math function won't be confused by using the angle numbers to calculate the sizes in the main driver code. I create my triangle object and give it a list of numbers. My constructor takes those numbers and immediately put them into the correct private variables so the triangle remembers its own sizes from the very start. Before the computation, I run my validation function to check if the angles add up to exactly 180 degrees. I put this here as a checker to make sure the shape is actually a triangle before the program proceeds. Once it passes, my Area() and Perimeter() functions take over. My perimeter function simply adds up the three side lengths, while my area function multiplies the base and height and cuts it in half. And lastly, my classification function checks and identifies the angles using a basic if-else condition as checker, it sees that one angle is bigger than 90 degrees and prints out that the triangle is obtuse. |

**Conclusion**

I learned how to prototype and use methods and what I'll put inside the main only and even it took me a while to finish this, I grasped how to use them but not in a such a complex way and how to and how to make my code clean and organized like the functions prototype it tells it to the compiler about it before being used, I learned to create objects and call methods and also noticed how my code become more cleaner when using it. Although it took me a while to understand and learn how to use this by practicing and noticing my mistakes several times just by my simple code to run, I actually did it and helped me correct my mistakes and understand the concepts better. I hope learn more about this before diving into another deeper topic to learn again for the next activity

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## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** _____

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** _____

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** _____

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

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